

ARPIT MAKKAR

ASPIRING SOFTWARE ENGINEER

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Summary

Computer Science student at VIT-AP (graduating 2027) who builds things that matter — from a 50-node AI-powered cyber deception network to a women's safety platform and a hackathon-winning expense tracker. Currently contributing to automotive software at Visteon Corporation as a Visteon Scholar, with prior experience as a Data Science Intern at Sabudh Foundation (STPI). Skilled across the full stack (MERN, FastAPI), cloud infrastructure (AWS, Oracle), and applied AI/ML. AWS Certified Cloud Architect, Oracle Database Professional, and Dataiku MLOps Practitioner — driven by the belief that the best code solves real problems.

Education

Vellore Institute of Technology, Amaravati, Andhra Pradesh, India

2023 - 2027

- B.Tech in Computer Science (Specialization in Data Analytics) | CGPA: 9.09 (Upto 5th Semester)

(expected)

Technical Skills

Languages: Java | Python | C | C++ | JavaScript

Frontend: ReactJS | HTML5 | CSS3 | Tailwind CSS

Backend: Node.js | Express.js | FastAPI | REST APIs | JWT Authentication

Databases: MySQL | MongoDB | Oracle Autonomous DB | Elasticsearch

Cloud & DevOps: AWS | Oracle Cloud | Docker | Docker Swarm | Git | GitHub | Vercel | Render

AI / ML & Other: Dataiku | Pandas | NumPy | Gemini LLM | Cohere LLM | Ollama (Mistral 7B) | ChromaDB | Firebase Studio | Postman | Qiskit

Work Experience

Visteon Scholar Intern – Visteon Corporation (Automotive Technology)

JAN 2026 – Present

- Visteon is a global automotive technology leader specializing in cockpit electronics, digital instrument clusters, infotainment systems, and connected vehicle software powering next-generation automotive experiences.
- Applying software engineering skills in an industry-grade automotive technology environment, working on engineering tasks alongside Visteon technologists, gaining hands-on exposure to production-grade automotive software engineering practices and industry development workflows.

Data Science Intern – Sabudh Foundation (STPI Initiative) – Remote, Mohali

[GitHub Repo](#)

JULY 2025 – Dec 2025

- Led design of a decision-support tool for EV infrastructure planning, integrating ML for demand forecasting and multi-objective optimization for site selection.
- Built Python data pipelines (pandas, NumPy) for large dataset cleaning and transformation; engineered composite feature sets including technical indicators to improve model accuracy.
- Researched Genetic Algorithms for hyperparameter optimization in market value prediction; presented findings in weekly mentor reviews, mirroring agile sprint cycles.

Web Development Intern - Google Developer Group (GDG), VIT-AP

[Certificate](#) [GitHub Repo](#)

NOV 2024 – Present

- Built the official GDG VIT-AP Feedback Page using ReactJS and Tailwind CSS — multi-step forms, state management, input validation, and responsive UI aligned with Figma designs and GDG branding.
- Awarded Certificate of Appreciation as Core Team Member (Web & App) for community contributions during 2024–2025.

Projects

Project Eidolon – AI-Powered Cyber Deception System (Automated Adaptive Minefield)

(2026)

Python | Docker Swarm | ELS Stack (Elasticsearch · Logstash · Kibana) | Mistral 7B (Ollama) | systemd

- Deployed a 50-node self-healing honeypot network using Docker Swarm; each container auto-generates a unique corporate identity (hostname, AWS honeytokens, SSH keys, fake DB credentials) via Python Faker — making every decoy server appear as a distinct real corporate machine to attackers.
- Built complete attack-intelligence pipeline: custom Python HTTP server → Logstash TCP → Elasticsearch time-series DB → live Kibana dashboard; a Python alert engine running as a systemd service dispatches Gmail SMTP breach alerts within 60 seconds of detection.
- Integrated locally-run Mistral 7B LLM (Ollama) to pre-generate unique convincing fake corporate page responses per node (wp-admin, .env, API endpoints); planted AWS honeytokens that beacon attacker location via CloudTrail if stolen.

SwarSmriti – AI Memory Voice Companion

(Google CodeClash 2025)

ReactJS | FastAPI | Gemini LLM | Firebase Studio | Render

[GitHub Repo](#) [Demo Video](#)

- Built with the vision of preserving human essence beyond life, SwarSmriti enables users to record meaningful memories and later engage in conversations with their AI self.
- Aims to offer emotional continuity, support grief healing, and safeguard personal legacies through intelligent, memory-aware interaction.
- Focused on creating a deeply human AI experience that bridges memory, voice, and presence.

Citadel Maps (City Sakhi) - AI-Powered Street Safety Heatmap

(CodeStorm 2025 Hackathon)

HTML | CSS | JavaScript | Google Maps API | Firebase Studio | Vercel

[GitHub Repo](#) [Demo Video](#)

- Co-developed a safety-focused navigation platform to enhance nighttime urban safety using community-driven data specifically for women.
- Engineered core features including real-time safety indicators, user-submitted hazard reporting, and instant alerts for nearby risks.
- Implemented a GIS-based user interface with AI-driven analytics to provide comprehensive safety ratings for pedestrian routes.
- Empowered users with data-driven insights to make informed navigation choices and foster community-led safety improvements.

Achievements

- Visteon Scholar Intern** — Selected for the competitive Visteon Scholar Program at Visteon Corporation (Jan 2026).
- Data Science Intern** — Sabudh Foundation, a STPI Government of India initiative (Jul – Dec 2025).
- Sabudh Foundation Data Science Batch-15 Performance** — A+ in DSA & Structured Query Language | A in Python Programming & Machine Learning | B+ in Natural Language Processing & Deep Learning.

[Certificate](#)

[Certificate](#)

Certifications

- Full Stack MERN Web Development (DELTA - APNA COLLEGE)**
- AWS Academy Graduate - Cloud Architecting & Cloud Foundations**
- Oracle Autonomous Database Cloud 2025 Certified Professional**
- Dataiku Certified:** MLOps Practitioner , ML Practitioner , Advanced Designer , Core Designer

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